SOLUTIONS

WORKSHEET: BASIC EXERCISES ON LAPLACIANS AND POISSON EQUATION

recall: $c_n = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} f(x) e^{-inx}$, $\|f\|_{L^2}^2 = \sum_{n=-\infty}^{\infty} |c_n|^2$

1. Let $f(x) = \sin(3x)$.

(a) Prove that $f \in H^1(-\pi, \pi)$ via the original definition.

$$\|f\|_{H^1}^2 = \int_{-\pi}^{\pi} |\sin(3x)|^2 dx + \int_{-\pi}^{\pi} |-3\cos(3x)|^2 dx$$

$$= \frac{1}{2} \int_{-\pi}^{\pi} 1 - \underbrace{\cancel{\cos(6x)}}_{\int = 0} dx + \frac{9}{2} \int_{-\pi}^{\pi} 1 + \underbrace{\cancel{\cos(6x)}}_{\int = 0} dx = \frac{1}{2} \cdot 2\pi + \frac{9}{2} \cdot 2\pi$$

$$= 10\pi \quad \therefore \boxed{\|f\|_{H^1} = \sqrt{10\pi}}$$

(b) Prove that $f \in H^1(S^1) = H^1_{per}(-\pi, \pi)$ via the Fourier series (summability-of-coefficients) definition.

$$c_n = \frac{1}{\sqrt{2\pi}} \int_{-\pi}^{\pi} \sin(3x) e^{-inx} dx = \frac{1}{2i\sqrt{2\pi}} \int_{-\pi}^{\pi} \left(e^{i3x} - e^{-i3x}\right) e^{inx} dx$$

$$= \frac{1}{2i\sqrt{2\pi}} \begin{cases} 0, & n \neq 3, -3 \\ +2\pi, & n = -3 \\ -2\pi, & n = +3 \end{cases} = \begin{cases} 0, & n \neq 3, -3 \\ -\sqrt{\frac{\pi}{2}}\, i, & n = -3 \\ +\sqrt{\frac{\pi}{2}}\, i, & n = +3 \end{cases}$$

$$\therefore \quad \|f\|_{H^1}^2 = \sum_{n=-\infty}^{\infty} (1 + n^2) |c_n|^2 = 10 \cdot \frac{\pi}{2} + 10 \cdot \frac{\pi}{2} = 10\pi \quad \therefore \boxed{\|f\|_{H^1} = \sqrt{10\pi}}$$

2. Let $f(x, y) = 6y(x - x^2) + 2(y - y^3)$ be the data on the unit square $\Omega = (0, 1)^2$.

(a) Show that $u(x, y) = (x - x^2)(y - y^3)$ solves the (strong form) of the homogeneous Dirichlet problem for the Poisson equation on Ω: $\quad -\nabla^2 u = f, \quad u = 0$ on $\partial\Omega$.

① $u|_{\partial\Omega} = 0$ since $x = 0, 1$ on $\partial\Omega$ or $y = 0, 1$ on $\partial\Omega$

② $-\nabla^2 u = -u_{xx} - u_{yy} = -(-2)(y - y^3) - (x - x^2)(-6y)$

$$= 6y(x - x^2) + 2(y - y^3) = f$$

(b) Describe in a few words, if you can, how I must have come up with this example.

I chose u first, with a product formula which hit zero at $x = 0, 1$ and $y = 0, 1$. Then I computed f as $-\nabla^2 u$.

method of manufactured solutions

3. Let $u(x,y) = \sin(\pi x)e^{\pi y}$ on the unit square $\Omega = (0,1)^2$.

$$\nabla u = \langle \pi\cos(\pi x)e^{\pi y},\ \pi\sin(\pi x)e^{\pi y} \rangle$$

(a) Compute the H^1 norm of u.

$$\|u\|_{H^1}^2 = \int_\Omega |u|^2 dx + \int_\Omega |\nabla u|^2 dx$$

$$= \int_0^1\int_0^1 \sin^2(\pi x)e^{2\pi y}\,dx\,dy + \int_0^1\int_0^1 \left(\pi^2\cos^2(\pi y) + \pi^2\sin^2(\pi y)\right)e^{2\pi y}\,dx\,dy$$

$$= \left(\int_0^1 \sin^2(\pi x)dx\right)\left(\int_0^1 e^{2\pi y}dy\right) + \pi^2 \cdot 1 \cdot \left(\int_0^1 e^{2\pi y}dy\right)$$

$$= \frac{1}{2}\cdot\frac{e^{2\pi}-1}{2\pi} + \pi^{\cancel{2}}\,\frac{e^{2\pi}-1}{2\cancel{\pi}}$$

$$= \left(\frac{1}{4\pi} + \frac{\pi}{2}\right)\left(e^{2\pi}-1\right) \qquad \therefore \boxed{\|u\|_{H_1} = \sqrt{\left(\frac{1}{4\pi}+\frac{\pi}{2}\right)\left(e^{2\pi}-1\right)}}$$

(b) Show that u solves the (strong form) of the Laplace equation on Ω.

$$\nabla^2 u = u_{xx} + u_{yy} = \left(\pi\cos(\pi x)e^{\pi y}\right)_x + \left(\pi\sin(\pi x)e^{\pi y}\right)_y$$

$$= -\pi^2\sin(\pi x)e^{\pi y} + \pi^2\sin(\pi x)e^{\pi y}$$

$$= 0\,\sin(\pi x)e^{\pi y} = 0$$

(c) Describe a boundary value problem, on Ω, which u satisfies.

several correct answers

1. $g_D = u\big|_{\partial\Omega}$
2. $g_N = \nabla u\big|_{\partial\Omega}\cdot\hat{n}$
3. do 1 and 2 on a partition of the boundary